

INMO(35th) – 2020

Time – 4 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
 - Ruler and compass allowed.
 - Answer all the questions.
 - All Questions carry equal marks. Maximum marks: 102.
 - Answer to each question should start on a new page. Clearly indicate the question number.
1. Let Γ_1 and Γ_2 be two circles of unequal radii with centres O_1 and O_2 respectively, in the plane intersecting in two distinct points A and B. Assume that the centre of each of the circle Γ_1 and Γ_2 is outside the other. The tangent to Γ_1 at B intersects Γ_2 again in C, different from B; the tangent to Γ_2 at B intersects Γ_1 again in D, different from B. The bisector of $\angle DAB$ and $\angle CAB$ meet Γ_1 and Γ_2 again in X and Y, respectively, different from A. Let P and Q be the circumcentres of triangle ACD and XAY respectively. Prove that PQ is the perpendicular bisector of the line segment O_1O_2 .
 2. Suppose $P(x)$ is a polynomial with real coefficients satisfying the condition $P(\cos\theta + \sin\theta) = P(\cos\theta - \sin\theta)$ for every real θ . Prove that $P(x)$ can be expressed in the form: $P(x) = a_0 + a_1(1-x^2)^2 + a_2(1-x^2)^4 + \dots + a_n(1-x^2)^{2n}$ for some real numbers $a_1, a_1, a_2, \dots, a_n$ and non-negative integer n.
 3. Let $X = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$. Let $S \subseteq X$ be such that any nonnegative integer n can be written as $p+q$ where the nonnegative integers p, q have all their digits in S. Find the smallest possible number of elements in S.
 4. Let $n \geq 3$ be an integer and let $1 < a_1 \leq a_2 \leq a_3 \leq \dots \leq a_n$ be n real numbers such that $a_1 + a_2 + a_3 + \dots + a_n = 2n$. Prove that $a_1 a_2 a_3 \dots a_{n-1} + a_1 a_2 a_3 \dots a_{n-2} + \dots + a_1 a_2 + a_1 + 2 \leq a_1 a_2 a_3 \dots a_n$.
 5. Infinitely many equidistant parallel lines are drawn in a plane. A positive integer $n \geq 3$ is called *frameable* if it is possible to draw a regular polygon with n sides all whose vertices lie on these lines and no line contains more than one vertex of a polygon.
 - (a) Show that 3, 4, 6 are *frameable*.
 - (b) Show that any integer $n \geq 7$ is not *frameable*.
 - (c) Determine whether 5 is *frameable*.
 6. A stromino is a 3×1 rectangle. Show that a 5×5 board divided into twenty five 1×1 squares cannot be covered by 16 strominos such that each stromino covers exactly three unit squares of the board and every unit square is covered by either one or two strominos. (A stromino can be placed either horizontally or vertically on the board.)

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